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SEMICONDUCTOR DEVICE WITH BACKSIDE SELF-ALIGNED POWER RAIL AND METHODS OF FORMING THE SAME

Abstract

Semiconductor device and the manufacturing method thereof are disclosed. An exemplary semiconductor device comprises a dielectric layer formed over a conductive feature; a semiconductor stack formed over the dielectric layer, wherein the semiconductor stack including semiconductor layers stacked up and separated from each other; a first metal gate structure and a second metal gate structure formed over a channel region of the semiconductor stack, wherein the first metal gate structure and the second metal gate structure wrap each of the semiconductor layers of the semiconductor stack; and a first epitaxial feature disposed between the first metal gate structure and the second metal gate structure over a first source/drain region of the semiconductor stack, wherein the first epitaxial feature extends through the dielectric layer and contacts the conductive feature.

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Background/Summary

PRIORITY DATA [0001] This application is a continuation application of U.S. patent application Ser. No. 18/588,727, filed Feb. 27, 2024, which is a continuation application of U.S. patent application Ser. No. 17/871,509, filed Jul. 22, 2022, which is a continuation application of U.S. patent application Ser. No. 17/199,629, filed Mar. 12, 2021, each of which is hereby incorporated herein by reference in its entirety.

BACKGROUND

[0002] The integrated circuit (IC) industry has experienced exponential growth. Multi-gate devices have been introduced to improve gate control by increasing gate-channel coupling and reducing off-state current. One such multi-gate device is a gate-all-around (GAA) device. A GAA device generally refers to any device having a gate structure, or portions thereof, formed on more than one side of a channel region (for example, surrounding a portion of the channel region). GAA transistors allow aggressive scaling down of transistors. However, such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advances to be realized, similar developments in IC processing and manufacturing are needed. For example, along with the scaling down of the device sizes, power lines (also referred to as power rails) are formed on the backside of the substrate. However, the existing power rail schemes face various challenges. For example, due to the process limitations, the existing power rail schemes has low density and high resistance, the isolation between the metal gate and the power line are very challenged, and the short issue from the source region to the drain region may also occurs. Therefore, the performance of the semiconductor is degraded, and improvements are needed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 illustrates a flowchart of an example method for making an example semiconductor device in accordance with some embodiments of the present disclosure.

[0005] FIGS. 2A-22A illustrate three-dimensional perspective views of the example semiconductor device at intermediate stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure.

[0006] FIGS. 2B-22B illustrate cross-sectional views of the semiconductor device along line A-A'

in the corresponding three-dimensional perspective views of the example semiconductor device at intermediate stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may comprise embodiments in which the first and second features are formed in direct contact, and may also comprise embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

[0008] In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Moreover, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may comprise embodiments in which the features are formed in direct contact, and may also comprise embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, “lower,” “upper,” “horizontal,” “vertical,” “above,” “over,” “below,” “beneath,” “up,” “down,” “top,” “bottom,” etc. as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features. Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm.

[0009] The present disclosure is generally related to semiconductor devices and the fabrication thereof, and more particularly to methods of fabricating field-effect transistors (FETs), such as gate-all-around FETs (GAA FETs).

[0010] In a GAA device, a channel region of a single device may comprise multiple layers of semiconductor material physically separated from one another. In some examples, a gate of the device is disposed above, alongside, and even between the semiconductor layers of the device. However, in a conventional GAA device, the power rail is disposed on the front side of the substrate (above the top surface of the substrate), the device performance is degraded due to the lower gate density and the higher resistance. Moving power rail from front side to backside is area benefited for GAA device. It can increase the metal line track number and improve the gate density. It can also increase the power rail width to lower the resistance. However, due to the process limitation and fabrication constraint, in a backside power rail scheme, overlay shifting between the via and the source/drain (S/D) feature is difficult to control which may cause short circuit between the metal gate and the power rail, and short circuit between source and drain regions.

[0011] The present disclosure provides a GAA device with a self-aligned backside power rail scheme. A bottom dielectric layer is applied to control the self-aligned via position. Thus, the via is controlled to only connect to the source region(s), thereby to mitigate the short circuit issues between the metal gate and the power rail, or between the source and the drain regions. Therefore, the reliability and the performance of the semiconductor device can be improved.

[0012] FIG. 1 illustrates a flow chart of a method **100** for making an example semiconductor device **200** (hereinafter, device **200**) in accordance with some embodiments of the present disclosure. Method **100** is merely an example and is not intended to limit the present disclosure

beyond what is explicitly recited in the claims. Additional operations can be performed before, during, and after method **100**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. Method **100** is described below in conjunction with other figures, which illustrate various three-dimensional and cross-sectional views of the device **200** during intermediate steps of method **100**. In particular, FIGS. **2A-22A** illustrate three-dimensional views of the device **200** at intermediate stages of the method **100** in accordance with some embodiments of the present disclosure. FIGS. **2B-22B** illustrate cross-sectional views of the device **200** taken along the plane A-A' shown in the corresponding three-dimensional views (that is, in an X-Z plane) at intermediate stages of the method **100** in accordance with some embodiments of the present disclosure.

[0013] Device **200** may be an intermediate device fabricated during processing of an integrated circuit (IC), or a portion thereof, that may comprise static random-access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as p-type FETs (PFETs), n-type FETs (NFETs), metal-oxide semiconductor field effect transistors (MOSFET), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, and/or other memory cells. Device **200** can be a portion of a core region (often referred to as a logic region), a memory region (such as a static random access memory (SRAM) region), an analog region, a peripheral region (often referred to as an input/output (I/O) region), a dummy region, other suitable region, or combinations thereof, of an integrated circuit (IC). In some embodiments, device **200** may be a portion of an IC chip, a system on chip (SoC), or portion thereof. The present disclosure is not limited to any particular number of devices or device regions, or to any particular device configurations. For example, though device **200** as illustrated is a three-dimensional GAA device, the present disclosure may also provide embodiments for fabricating fin-like FETs (FinFETs) or planar FET devices.

[0014] Referring to FIGS. **1**, **2A** and **2B**, at operation **105**, an initial semiconductor structure of device **200** is formed. As depicted in FIGS. **2A** and **2B**, device **200** comprises a substrate **202**. In the depicted embodiment, the substrate **202** is a bulk silicon substrate. Alternatively or additionally, the substrate **202** includes another single crystalline semiconductor, such as germanium; a compound semiconductor; an alloy semiconductor; or combinations thereof. Alternatively, the substrate **202** is a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. The substrate **202** may be doped with different dopants to form various doped regions therein. For example, the substrate **202** may include PFET region comprising n- type doped substrate regions (such as n-well) and NFET region comprising p-type doped substrate regions (such as p-well).

[0015] The device **200** includes alternating semiconductor layers formed over the substrate **202**, such as semiconductor layers **210A** including a first semiconductor material and semiconductor layers **210B** including a second semiconductor material that is different from the first semiconductor material. The different semiconductor materials of the semiconductor layers **210A** and **210B** have different oxidation rates and/or different etch selectivity. In some embodiments, the first semiconductor material of the semiconductor layers **210A** is the same as the substrate **202**. For example, the semiconductor layers **210A** comprise silicon (Si, like the substrate **202**), and the semiconductor layers **210B** comprise silicon germanium (SiGe). Thus, alternating Si/SiGe/Si/SiGe/Si/ . . . layers are arranged from bottom to top. In some embodiments, the material of the top semiconductor layer may or may not be the same as the bottom semiconductor layer. In some embodiments, the semiconductor layers **210A** may be undoped or substantially dopant-free. In other words, no intentional doping is performed when forming the semiconductor layers **210A**. In some other embodiments, the semiconductor layers **210A** may be doped with a p-type dopant or an n-type dopant. The number of the semiconductor layers **210A** and **210B** depends on the design

requirements of device **200**. For example, it may comprise one to ten layers of semiconductor layers **210A** or **210B** each. In some embodiments, different semiconductor layers **210A** and **210B** have the same thickness in the Z-direction. In some other embodiments, different semiconductor layers **210A** and **210B** have different thicknesses. In some embodiments, the semiconductor layers **210A** and/or **210B** are formed by suitable epitaxy process. For example, semiconductor layers comprising SiGe and Si are formed alternately over the substrate **202** by a molecular beam epitaxy (MBE) process, a chemical vapor deposition (CVD) process, such as a metal organic CVD (MOCVD) process, and/or other suitable epitaxial growth processes.

[0016] Referring to FIGS. **2A** and **2B**, a semiconductor layer **204** is formed in the substrate **202** and separating the substrate **202** into a top portion **202T** and a bottom portion **202B**. The semiconductor layer **204** will serve as an etch stop layer (ESL) when removing the substrate **202** to form the power rail on the backside, and thus a material of the semiconductor layer **204** is different from that of the substrate **202** and can provide a different oxidation rate and/or etching selectivity than the substrate **202**. In some embodiment, the semiconductor layer **204** comprises a semiconductor material including silicon (Si), germanium (Ge), silicon germanium (SiGe), other semiconductor material, or combinations thereof. For example, in the depicted embodiment, the material of the semiconductor layer **204** is SiGe with a Ge concentration of more than about 15%. In some embodiments, the semiconductor layer **204** is formed by an implantation process. For example, the substrate **202** is a silicon substrate, germanium ions are implanted into the substrate **202** with a desired dose and energy to be located beneath a surface silicon layer in the substrate **202**, such that the semiconductor layer **204** including SiGe is formed between the top portion **202T** and the bottom portion **202B** of the substrate **202**. In some embodiments, the implantation is performed before or after the formation of the alternating semiconductor layers **210A** and **210B**. As depicted in FIG. **2B**, in some embodiments, the semiconductor layer **204** has a thickness **H1** of about 10 nm to about 50 nm, such that it can effectively work as an ESL while not taking too much space in the substrate **202**. In the following description, the semiconductor layer **204** is considered as a portion of the substrate **202**.

[0017] A sacrificial layer **206** is formed between the substrate **202** and the lowermost semiconductor layer **210A**. The sacrificial layer **206** is going to be removed during the S/D trench formation and will be replaced by a bottom dielectric layer **230** (FIGS. **10A** and **10B**) serving the self-align function. Thus, a material of the sacrificial layer **206** is different from the substrate **202** and the semiconductor layers **210A** and **210B** to provide different oxidation rates and/or etching selectivity in the later etching processes. In some embodiment, the sacrificial layer **206** comprises a semiconductor material including silicon (Si), germanium (Ge), silicon germanium (SiGe), other semiconductor material, or combinations thereof. In the depicted embodiment, the material of the sacrificial layer **206** is also SiGe but with a lower Ge concentration than the SiGe material of the semiconductor layers **210B**. In some embodiments, a Ge concentration of the sacrificial layer **206** is about 10% to about 35%, while a Ge concentration of the semiconductor layers **210B** is about 20% to about 50%. In some embodiment, the sacrificial layer **206** is also formed by an implantation process, similar as the implantation process to form the semiconductor layer **204**. The sacrificial layer **206** may be formed before or after the formation of the alternating semiconductor layers **210A** and **210B** and/or the semiconductor layer **204**. As depicted in FIG. **2B**, in some embodiments, the sacrificial layer **206** has a thickness **H2** of about 6 nm to about 20 nm, such that the later formed bottom dielectric layer **230** has the thickness of about 6 nm to about 20 nm which can provide the effective self-align function.

[0018] Now referring to FIGS. **1**, **3A**, **3B**, **4A** and **4B**, at operation **110**, the alternating semiconductor layers **210A** and **210B** are patterned to form semiconductor stacks **210** (hereinafter the stacks **210**), and dummy gate structures **220** are formed over the stacks **210**. Referring to FIGS. **3A** and **3B**, a photoresist and an etching process may be performed to the semiconductor layers **210A** and **210B** to form the stacks **210** in fin-shapes as illustrated. For example, first, a patterned

photoresist mask is formed over the semiconductor device. The patterned photoresist mask exposes the fin positions according to the design requirement of device **200**. Subsequently, one or more etching processes are performed using the patterned photoresist mask to form the stacks **210**. The etching process includes dry etching, wet etching, other suitable etching process, or combinations thereof. In the depicted embodiment, the one or more etching processes remove, along the patterned photoresist mask, the semiconductor layers **210A**, **210B**, as well as the sacrificial layer **206** and a part of the top portion **202T** of the substrate **202** to form trenches between the fin-shape stacks **210**.

[0019] Thereafter, still referring to FIGS. **3A** and **3B**, an isolation structure **208** is formed in the trenches between the stacks **210** to separate and isolate the active regions of device **200**. In some embodiments, one or more dielectric materials, such as silicon dioxide (SiO₂) and/or silicon nitride (SiN), is deposited over the substrate **202** along sidewalls of the stack **210**. The dielectric material may be deposited by CVD (such as plasma enhanced CVD (PECVD)), physical vapor deposition (PVD), thermal oxidation, or other techniques. Subsequently, the dielectric material is recessed (for example, by etching and/or chemical mechanical polishing (CMP)) to form the isolation structure **208**. In some embodiments, a top surface of the isolation structure **208** is substantially coplanar with or below a bottom surface of the sacrificial layer **206**, as depicted in FIG. **3A**. Subsequently, a liner layer **212** is formed over the isolation structure **208**, along the sidewalls and over the top surfaces of the stacks **210**. In some embodiments, the liner layer **212** may include a dielectric material such as SiO₂, SiN, SiON, and/or other dielectric material. In some embodiments, the liner layer **212** is formed by ALD, CVD, PVD, other deposition process, or combinations thereof.

[0020] Then, referring to FIGS. **4A** and **4B**, dummy gate structures **220-1** and **220-2** (both referred to as dummy gate structures **220**) are formed over the stacks **210**. Each dummy gate structure **220** serves as a placeholder for subsequently forming a metal gate structure. In some embodiments, the dummy gate structures **220** extend along the Y-direction and traverse respective stacks **210**. The dummy gate structures **220** cover the channel regions of the stacks **210** which interpose the source regions and the drain regions (both referred to as S/D regions). Each of the dummy gate structures **220** may include various dummy layers. For example, a dummy gate electrode **214** (including polysilicon), one or more hard mask layers **216** and **218** (including a dielectric material such as SiN, silicon carbonitride (SiCN), SiO₂, etc.), and/or other suitable layers. The dummy gate structures **220** are formed by deposition processes, lithography processes, etching processes, other suitable processes, or combinations thereof. For example, different dummy layers are deposited over the stacks **210**. A lithography process is then performed to form a mask covering the channel regions of the stacks **210**. Thereafter, the different dummy layers are etched using the lithography mask to form the dummy gate structures **220**. And, the lithography mask is removed using any proper method.

[0021] Gate spacers **222** are formed along the sidewalls of the dummy gate structures **220** over the stack **210**. In some embodiments, the gate spacers **222** comprise a dielectric material, such as SiO₂, SiN, silicon oxynitride (SiON), silicon carbide (SiC), other dielectric material, or a combination thereof. The gate spacers **222** are formed by any suitable process(es). For example, first, a spacer layer comprising the dielectric material is deposited (for example, by atomic layer deposition (ALD), CVD, PVD, or other proper process) over the isolation structure **208** and along sidewalls and over top surfaces of the dummy gate structures **220**. Subsequently, the spacer layer is anisotropically etched to remove the portions in the X-Y plane (the plane in which the top surface of the substrate **202** is). The remaining portions of the spacer layer then become the gate spacers **222**.

[0022] Thereafter, S/D regions of the stack **210** may be recessed along sidewalls of the gate spacers **222** to form S/D trenches **224S** and **224D** (all referred to as S/D trenches **224**) in the S/D regions of the stacks **210**. In the depicted embodiment, a source region is between the gate structures **220-1** and **220-2** and drain regions are along sidewalls of the dummy gate structures **220-1** and **220-2**.

away from each other. Thus, the source trench **224S** is a source trench over the source region of the stack **210** (i.e. source trench **224S**); and the drain trenches **224D** are drain trenches over the drain region of the stack **210** (i.e. drain trench **224D**). In some other embodiments, the region between the gate structures **220-1** and **220-2** may be a drain region, thus the trench formed therebetween is a drain trench; and the regions along sidewalls of the dummy gate structures **210-1** and **210-2** away from each other are source regions, thus the trenches formed there are source trenches. In some embodiments, the S/D regions of the stack **210** are recessed by a S/D etching process performed along the gate spacers **222** to form the S/D trenches **224**. The S/D etching process may be a dry etch (such as a reactive ion etching (RIE) process), a wet etch, or combinations thereof. The duration of the S/D etching process is controlled such that the sidewalls of each semiconductor layers **210A** and **210B** are exposed in the S/D trenches. In some other embodiments, the S/D etching process stops on the lowermost semiconductor layer **210A**, such that the lowermost semiconductor layer **210A** is not substantially etched, as depicted in FIG. **4B**.

[0023] Now referring to FIGS. **1**, **5A**, **5B**, **6A** and **6B**, at operation **115**, the source trench **224S** between the dummy gate structures **220-1** and **220-2** is further recessed, while the drain trenches **224D** are substantially unaffected. Referring to FIGS. **5A** and **5B**, a patterned photoresist mask **226** is formed over the device **200**. The patterned photoresist mask **226** expose the source trench **224S** between the dummy gate structures **220-1** and **220-2** and fills in the drain trenches **224D** along sidewalls of the dummy gate structures **220-1** and **220-2** away from each other. Referring to FIGS. **6A** and **6B**, a further S/D etching process is performed, using the patterned photoresist mask **226** as a mask, such that the source trench **224S** is further recessed while the drain trenches **224D** remain substantially unchanged. As depicted in FIG. **6B**, a bottom surface of the source trench **224S** is below a bottom surface of the sacrificial layer **206**, while a bottom surface of the drain trenches **224D** is above a top surface of the sacrificial layer **206**. In other words, the sacrificial layer **206** is separated in the source region between the dummy gate structures **220-1** and **220-2**, and sidewalls of the sacrificial layer **206** are exposed in the further recessed source trench **224S**. The patterned photoresist mask **226** is then removed after the further recessing of the source trench **224S**.

[0024] Now referring to FIGS. **1**, **7A**, **7B**, **8A**, and **8B**, at operation **120**, inner spacers **228** are formed between the edges of the semiconductor layers **210A**. Referring to FIGS. **7A** and **7B**, the exposed portions (edges) of the semiconductor layers **210B** in the S/D trenches **224** are selectively removed by a suitable etching process to form gaps between the semiconductor layers **210A**. In other words, edges of the semiconductor layers **210A** are suspended in the S/D trenches **224**. Due to the different oxidation rates and/or etching selectivities of the materials of the semiconductor layers **210A** (for example, Si), **210B** (for example, SiGe with a higher Ge concentration), and the sacrificial layer **206** (for example, SiGe with a lower Ge concentration), only exposed portions (edges) of the semiconductor layers **210B** are removed, while the semiconductor layers **210A** and the sacrificial layer **206** remain substantially unchanged. In some embodiments, the selective removal of the exposed portions of the semiconductor layers **210B** may include an oxidation process followed by a selective etching process. For example, the edge portions of the semiconductor layers **210B** are first selectively oxidized to include a material of SiGeO. Then, a selective etching process is performed to remove the SiGeO with a suitable etchant such as ammonium hydroxide (NH₄OH) or hydro fluoride (HF). The duration of the oxidation process and the selective etching process can be controlled such that only edge portions of the semiconductor layers **210B** are selectively removed. In some embodiments, when removing the edge portions of the semiconductor layers **210B**, portions of the bottom semiconductor layers **210A** exposed in the drain trenches **224D** may also be removed.

[0025] Subsequently, referring to FIGS. **8A** and **8B**, inner spacers **228** are formed to fill in the gaps between the semiconductor layers **210A**. The inner spacers **228** comprise a dielectric material that is similar to the material of the gate spacers **222**, such as SiO, SiN, SiON, SiC, or combinations thereof. The dielectric material of the inner spacers may be deposited in the S/D trenches **224** and

in the gaps between the edges of the semiconductor layers **210A** by CVD, PVD, ALD, or combinations thereof. Extra dielectric material is then removed along sidewalls of the gate spacers **222** until the sidewalls of the semiconductor layers **210A** are exposed in the S/D trenches **224**. [0026] Now referring to FIGS. **1**, **9A**, **9B**, **10A**, and **10B**, at operation **125**, the sacrificial layer **206** is removed and replaced with a bottom dielectric layer **230**. Referring to FIGS. **9A** and **9B**, since the sidewalls of the sacrificial layer **206** are exposed in the source trench **224S**, the sacrificial layer **206** can be selectively removed from the source trench **224S**. As mentioned above, the material of the sacrificial layer **206** can provide different oxidation rate and/or etching selectivity than the semiconductor layers **210A** and **210B**, and the semiconductor layers **210B** are protected by the inner spacers **228**, thus the selective removing process only removes the sacrificial layer **206** and forms a gap between the substrate **202** and the stacks **210**. The selective removing process is similar as the process(es) to remove the edge portions of the semiconductor layers **210B**. [0027] Thereafter, referring to FIGS. **10A** and **10B**, still at operation **125**, the bottom dielectric layer **230** is formed in the gap between the substrate **202** and the stacks **210** to replace the removed sacrificial layer **206**. In some embodiments, the bottom dielectric layer **230** includes a material such as SiO, SiN, SiCN, silicon oxycarbonitride (SiOCN), metal oxide, other dielectric material, or combinations thereof. In some embodiments, the dielectric material of the bottom dielectric layer **230** is first deposited, from the opening of the source trench **224S**, in the gap and in the bottom portion of the source trench **224S** by a suitable deposition process, for example, CVD, PVD, ALD, other deposition process, or a combination thereof. Then, an anisotropic etching is performed to remove the dielectric material in the bottom portion of the source trench **224S**. The anisotropic etching includes dry etching, wet etching, or combinations thereof. The remained dielectric material after the anisotropic etching forms the bottom dielectric layer **230**. As depicted in FIG. **10B**, sidewalls of the bottom dielectric layer **230** are exposed in the source trench **224S**. In other words, the bottom dielectric layer **230** is separated in the source region of the stacks **210**, but is extended under the drain regions of the stacks **210**. Therefore, the later formed epitaxial feature **232S** in the source region may contact the later formed via **250** through the bottom dielectric layer **230** in a self-aligned manner (FIG. **20B**). In some embodiments, a thickness of the bottom dielectric layer **230** is the same as the thickness H2 of the sacrificial layer **206**, which is about 6 nm to about 20 nm.

[0028] Now referring to FIGS. **1**, **11A**, and **11B**, at operation **130**, epitaxial S/D features **232S** and **232D** (all referred to as epitaxial S/D features **232**) are formed in the S/D trenches **224**. In the depicted embodiment, the epitaxial source feature **232S** is epitaxial grown in the source trench **224S** and the epitaxial drain features **232D** are epitaxial grown in the drain trenches **224D**, respectively. In some embodiments, the epitaxial S/D features **232** may comprise a semiconductor material such as Si or Ge; a compound semiconductor such as SiGe, SiC, gallium arsenide (GaAs), etc.; an alloy semiconductor; or combinations thereof. An epitaxy process may be implemented to epitaxially grow the S/D features **232**. The epitaxy process may comprise CVD deposition (for example, vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), low-pressure CVD (LPCVD), and/or plasma-enhanced (PECVD)), molecular beam epitaxy (MBE), other suitable selective epitaxial growth (SEG) processes, or combinations thereof. The epitaxial S/D features **232** may be doped with n-type dopants and/or p-type dopants. In some embodiments, the epitaxial S/D features **232** may comprise multiple epitaxial semiconductor layers, and different epitaxial semiconductor layers are different in amount of dopant comprised therein. As depicted in FIG. **11B**, the epitaxial source feature **232S** has a bottom surface below the bottom surface of the bottom dielectric layer **230** and contacts the substrate **202**. Referring to FIG. **11B**, the bottom surface of the epitaxial source feature **232S** has a width W1. In some embodiments, the width W1 is more than about 10 nm. The epitaxial drain features **232D** have bottom surfaces contacting the bottom dielectric layer **230**. In other words, the epitaxial source feature **232S** punches through the bottom dielectric layer **230** and connect to the top portion **202T** of the substrate **202**, while the epitaxial

drain features **232D** are separated from the substrate **202** by the bottom dielectric layer **230**.

[0029] Now referring to FIGS. **1**, **12A**, **12B**, **13A**, **13B**, **14A**, and **14B**, at operation **135**, a metal gate replacement process is performed to replace the dummy gate structures **220-1** and **220-2** with metal gate structures **240-1** and **240-2**, respectively. Both metal gate structures **240-1** and **240-2** are referred to as metal gate structures **240**. The metal gate replacement process includes various processing steps. For example, first, an interlayer dielectric (ILD) layer **234** is formed over the substrate **202**. As illustrated in FIGB. **12A** and **12B**, the ILD layer **234** is disposed along the gate spacers **222** and over the isolation structure **208** and the epitaxial S/D features **232**. In some embodiments, an etch stop layer (ESL) **236** is deposited before the ILD layer **234** and between the ILD layer **234** and the isolation structure **208** and between the ILD layer **234** and the epitaxial S/D features **232**. In some embodiments, the ESL **236** includes a dielectric material such as SiO, SiON, SiN, SiCN, SiOC, SiOCN, other suitable materials, or combinations thereof. In some embodiments, the ILD layer **234** comprises a low-k ($K < 3.9$) dielectric material, such as tetraethylorthosilicate (TEOS), un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fluorosilicate glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), other suitable dielectric materials, or combinations thereof. The ESL **236** and the ILD layer **234** may be formed by deposition processes such as CVD, flowable CVD (FCVD), spin-on-glass (SOG), other suitable methods, or combinations thereof.

[0030] Thereafter, the dummy gate structures **220** are removed to form gate trenches **238** exposing the channel regions of the stacks **210**. In some embodiments, removing the dummy gate structures **220** comprises one or more etching processes, such as wet etching, dry etching, reactive-ion etching (RIE), or other etching techniques. In some embodiments, top portions of the ESL **236**, the ILD layer **234**, and the spacer **222** are also removed at operation **135**. The semiconductor layers **210A** and **210B** are then exposed in the gate trenches **238**.

[0031] Subsequently, referring to FIGS. **13A** and **13B**, the semiconductor layers **210B** are selectively removed from the gate trenches **238**. Due to the different materials of the semiconductor layers **210A** and **210B**, the semiconductor layers **210B** are removed by a selective oxidation/etching process similar as those to remove the edge portions of the semiconductor layers **210B** or the sacrificial layer **206**. In some embodiments, the semiconductor layers **210A** are slightly etched or not etched during the operation **135**. Thereby, the semiconductor layers **210A** are suspended in the channel region of the stacks **210** and stacked up along a direction (Z-direction) generally perpendicular to a top surface of the substrate **202** (X-Y plane). The suspended semiconductor layers **210A** are also referred to as channel semiconductor layers **210A**.

[0032] Then, referring to FIGS. **14A** and **14B**, metal gate structures **240** are formed in the channel regions of the stacks **210**. The metal gate structures **240** wrap each of the suspended semiconductor layers **210A**. And each metal gate structure **240** includes an interfacial layer **242**, a gate dielectric layer **244**, a metal gate electrode **246**, and other metal gate layers. In some embodiments, the interfacial layer **242** includes any suitable material, for example, SiO. The gate dielectric layer **244** includes a high-k ($K > 3.9$) dielectric material, such as HfO₂, HfSiO, HfSiO₄, HfSiON, HfLaO, HfTaO, HfTiO, HfZrO, HfAlO_x, ZrO, ZrO₂, ZrSiO₂, AlO, AlSiO, Al₂O₃, TiO, TiO₂, LaO, LaSiO, Ta₂O₃, Ta₂O₅, Y₂O₃, SrTiO₃, BaZrO, BaTiO₃ (BTO), (Ba,Sr)TiO₃ (BST), Si₃N₄, hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, other suitable high-k dielectric material, or combinations thereof. In some embodiments, the interfacial layer **242** and the gate dielectric layer **244** are deposited by CVD, PVD, ALD, and/or other suitable method. The metal gate electrodes **246** are then filled in the spaces between the gate dielectric layer **244**. Each metal gate electrode **246** includes one or more work function metal (WFM) layers and a bulk metal. The WFM layer is configured to tune a work function of its corresponding transistor to achieve a desired threshold voltage V_t. And, the bulk metal is configured to serve as the main conductive portion of the functional gate structure. In some embodiments, the material of the WFM layer may include TiAl, TiAlC, TaAlC, TiAlN, TiN, TSN, TaN, WCN, Mo, other materials, or combinations thereof. The

bulk metal may include Al, W, Cu, or combinations thereof. The various layers of the metal gate electrodes **246** may be formed by any suitable method, such as CVD, ALD, PVD, plating, chemical oxidation, thermal oxidation, other suitable methods, or combinations thereof. Thereafter, one or more polishing processes (for example, CMP) are applied to remove any excess conductive materials and planarize the top surface of the device **200**.

[0033] Now referring to FIGS. **1**, **15A**, and **15B**, at operation **140**, various interconnection structures **247** are formed over the top (i.e. the front side) of the device **200**. In some embodiments, the various interconnection structures **247** may include dielectric ESL, ILD layers, conductive S/D contacts, vias, metal lines, and/or other structures, configured to connect the various features to form a functional circuit that may include one or more multi-gate devices. The interconnection structures **247** are simplified in the later drawings.

[0034] Now referring to FIGS. **1**, **16A**, **16B**, **17A**, **17B**, **18A**, and **18B**, at operation **145**, portions of the substrate **202** below the isolation structure **208** are removed from the bottom (i.e. the backside) of the device **200**. For the convenience of discussion, FIGS. **15A** and **15B** and the following figures show the device **200** in an upside-down direction. The removal of the portions of the substrate **202** includes various steps. For example, referring to FIGS. **16A** and **16B**, the bottom portion **202B** of the substrate **202** is removed until the semiconductor layer **204** is exposed (i.e. the semiconductor layer **204** is served as an ESL). In some embodiments, the bottom portion **202B** of the substrate **202** is removed by a CMP or other planarization process. In some further embodiments, the removal of the bottom portion **202B** of the substrate **202** involves an etching process, such as wet etching, dry etching, other etching process, or a combination thereof.

[0035] Thereafter, referring to FIGS. **17A** and **17B**, the semiconductor layer **204** is selectively removed since it includes a material (for example, SiGe with Ge concentration of more than about 15%) has a different selectivity from the substrate **202**. The selective removal of the semiconductor layer **204** is similar as the aforementioned selective oxidation/etching process to remove the semiconductor layers **210B** and the sacrificial layer **206**.

[0036] Thereafter, referring to FIGS. **18A** and **18B**, the top portion **202T** of the substrate **202** below the isolation structure **208** is removed to expose the isolation structure **208** from the bottom of the device **200**. Similar as the removal of the bottom portion **202B**, the top portion **202T** of the substrate **202** may be removed by a planarization process (for example, CMP) and/or an etching process (for example, wet etching, dry etching, or combinations thereof).

[0037] Now referring to FIGS. **1**, **19A** and **19B**, at operation **150**, the remained portions of the substrate **202** between the isolation structure **208** are selectively removed to form via trenches **248**. In some embodiments, before removing the remained portions of the substrate **202**, a lithography hard mask is formed to cover the electrostatic discharge (ESD) region(s) of the device **200**, such that the following operations including operations **150**, **155**, and **160** are only applied to the non-ESD region(s) of the device **200**. The substrate **202** includes a material having a different selectivity from the isolation structure **208**, the bottom dielectric layer **230**, and the epitaxial S/D features **232**, thus the remained portions of the substrate **202** can be selectively removed, for example, by a selective etching process. Therefore, via trenches **248** are formed between the isolation structure **208**. As depicted in FIG. **19A**, the bottom dielectric layer **230** and the bottom surface of the epitaxial source feature **232S** are exposed from the via trenches **248**. Referring to FIG. **19B**, the width of the exposed bottom surface of the epitaxial source feature **232S** is **W1**, which is more than about 10 nm. The bottom surfaces of the metal gate structures **240** and the epitaxial drain features **232D** in the drain regions are blocked by the bottom dielectric layer **230** and thus are not exposed. Therefore, the bottom dielectric layer **230** can serve as a self-align layer to decide the contacting position of the later formed power rail vias **250** and the epitaxial source feature **232S**, i.e. the power rail vias **250** only contact the epitaxial S/D feature in the source region (epitaxial source feature **232S**), not those in the drain region (epitaxial drain features **232D**) and not the metal gate structures **240**.

[0038] Now referring to FIGS. 1, 20A and 20B, at operation 155, power rail vias 250 are formed in the via trenches 248. In some embodiments, the power rail vias 250 includes a conductive material such as W, Ru, Co, Cu, Mo, Ni, Ti, TiN, Ta, TaN, other conductive material, or combinations thereof. In some embodiments, the conductive material of the power rail vias 250 are deposited by any suitable process in the via trenches 248 and contacting the bottom dielectric layer 230. And a planarization process (for example, CMP) is performed to remove the excess conductive material and expose the bottom dielectric layer 230. Thereby, the remained conductive material forms the vias 250. In some embodiments, a silicide layer 252 is formed between the epitaxial source feature 232S and the power rail vias 250 to reduce the parasitic resistance therebetween. For example, an annealing process may be performed to form the silicide layer 252 between the vias 250 and the epitaxial source feature 232S. The heat causes the constituents of the epitaxial source feature 232S to react with the power rail vias 250, thus the silicide layer 252 may include the conductive material of the power rail vias 250 and the constituent of the epitaxial source feature 232S, for example, TiSi, CoSi, MoSi, and etc. Referring to FIG. 20A, a width of the contact portion between the epitaxial source feature 232S and the power rail via 250 is W1, which is more than about 10 nm.

[0039] Now referring to FIGS. 1, 21A and 21B, at operation 160, power rail 260 is formed on the bottom (i.e. the backside) of the device 200. In some embodiments, the power rail 260 includes a conductive material such as Cu, Al, Co, W, Ti, Ta, Ru, other metal material, or combinations thereof. The power rail 260 may be formed by any suitable method, such as CVD, ALD, PVD, plating, chemical oxidation, thermal oxidation, other suitable methods, or combinations thereof. And, the shape of the power rail 260 can be patterned via a photoresist and/or etching process according to the design requirements of the device 200.

[0040] Now referring to FIGS. 1, 22A and 22B, at operation 165, other backside contact features 270 are formed such that the epitaxial source feature 232S can connect to the corresponding power supply (for example, Vdd) through the self-aligned power rail vias 250 and the power rail 260.

[0041] In a convention power rail scheme, due to the process limitation and fabrication constraint, overlay shifting issues may happen during the formation of the power rail vias, thus short circuit issues may occur between the power rail and the metal gate, and/or between the source and drain regions. However, in the present disclosure, a bottom dielectric layer is formed under the channel region and extends to the S/D regions. The bottom dielectric layer leaves an opening exposing only one type of the epitaxial S/D features (for example, the source feature), thereby power rail via can contact the exposed type of the S/D features in a self-aligned manner. Thus, the short circuit issues can be mitigated.

[0042] Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to an integrated circuit and a formation process thereof. For example, embodiments of the present disclosure provide a semiconductor device including a bottom dielectric layer below the channel regions and extends below one type of the S/D regions (for example, the drain regions). The bottom dielectric layer provides an opening at the bottom of one type of the S/D features (for example, the source features) and isolates the other type of the S/D features (for example, the drain features) and the metal gate from the backside power rails. Thus, the power rails only contact the exposed S/D features through the self-aligned vias and the short circuits issues between the power rails and the metal gate and between the source features and the drain features can be mitigated. Therefore, the reliability and the performance of the semiconductor device is improved.

[0043] The present disclosure provides for many different embodiments. Semiconductor device having backside self-aligned power rail and methods of fabrication thereof are disclosed herein. An exemplary semiconductor device comprises a dielectric layer formed over a conductive feature; a semiconductor stack formed over the dielectric layer, wherein the semiconductor stack including semiconductor layers stacked up and separated from each other; a first metal gate structure and a

second metal gate structure formed over a channel region of the semiconductor stack, wherein the first metal gate structure and the second metal gate structure wrap each of the semiconductor layers of the semiconductor stack; and a first epitaxial feature disposed between the first metal gate structure and the second metal gate structure over a first source/drain region of the semiconductor stack, wherein the first epitaxial feature extends through the dielectric layer and contacts the conductive feature

[0044] In some embodiments, the exemplary semiconductor device further comprises a power rail formed below the conductive feature, wherein the power rail electrically contacts the first epitaxial feature via the conductive feature.

[0045] In some embodiments, the exemplary semiconductor device further comprises a second epitaxial feature disposed over a second source/drain region of the semiconductor stack, wherein a bottom surface of the second epitaxial feature is over a top surface of the dielectric layer.

[0046] In some embodiments, the exemplary semiconductor device further comprises an isolation feature disposed along sidewalls of the conductive feature and separating active regions of the semiconductor device.

[0047] In some embodiments, a thickness of the dielectric layer is about 6 nm to about 20 nm. In some embodiments, a width of a contact portion between the conductive feature and the first epitaxial feature is more than about 10 nm. In some embodiments, the first epitaxial feature includes a silicide layer formed at a bottom surface of the first epitaxial feature contacting the conductive feature.

[0048] A method of forming a semiconductor device comprises forming a semiconductor stack over a substrate, wherein the semiconductor stack includes a sacrificial layer, and first semiconductor layers and second semiconductor layers alternately stacked up over the sacrificial layer, and each of a material of the first semiconductor layers, a material of the second semiconductor layers, and a material of the sacrificial layer provides a different etching selectivity; forming a first dummy gate structure and a second dummy gate structure over the semiconductor stack; forming a first source/drain (S/D) trench between the first dummy gate structure and the second dummy gate structure, wherein the first S/D trench extends below the sacrificial layer; replacing the sacrificial layer with a dielectric layer; and epitaxially growing a first epitaxial feature in the first S/D trench, wherein a bottom portion of the first epitaxial feature separates the dielectric layer between the first dummy gate structure and the second dummy gate structure.

[0049] In some embodiments, the exemplary method further comprises forming an isolation feature around a lower portion of the semiconductor stack to isolate active regions of the semiconductor device; selectively removing the second semiconductor layers; replacing the first dummy gate structure with a first metal gate structure and replacing the second dummy gate structure a second metal gate structure, wherein the first metal gate structure and the second metal gate structure wrap each of the first semiconductor layers; removing the substrate and the lower portion of the semiconductor stack to form an opening in the isolation feature, wherein the dielectric layer and the bottom portion of the first epitaxial feature are exposed from the opening; and forming a conductive feature below the dielectric layer and the bottom portion of the first epitaxial feature in the opening. In some embodiments, the exemplary method further comprises forming a power rail below and contacting the conductive feature. In some embodiments, a top surface of the isolation feature is below a bottom surface of the sacrificial layer.

[0050] In some embodiments, the exemplary method further comprises depositing an etch stop layer (ESL) in the substrate below the sacrificial layer, wherein the ESL separates the substrate to a first portion above the ESL and below the isolation feature and a second portion below the ESL; wherein the removing the substrate and the lower portion of the semiconductor stack includes removing the second portion of the substrate to expose the ESL; selectively removing the ESL; removing the first portion of the substrate to expose the isolation feature; and selectively removing the lower portion of the semiconductor stack to form the opening in the isolation feature.

[0051] In some embodiments, the forming the first S/D trench between the first dummy gate structure and the second dummy gate structure includes removing a portion of the semiconductor stack along sidewalls of the first dummy gate structure and the second dummy gate structure to form a first S/D trench between the first dummy gate structure and the second dummy gate structure; and further recessing the first S/D trench such that a bottom surface of the first S/D trench is below a bottom surface of the sacrificial layer. In some embodiments, the exemplary method further comprises removing a portion of the semiconductor stack along a sidewall of the second dummy gate structure away from the first dummy gate structure to form a second S/D trench, wherein a bottom surface of the second S/D trench is above a top surface of the sacrificial layer; and epitaxially growing a second epitaxial feature in the second S/D trench, wherein a bottom surface of the second epitaxial feature is above a top surface of the sacrificial layer.

[0052] In some embodiments, the replacing the sacrificial layer with the dielectric layer includes selectively removing the sacrificial layer from the first S/D trench to form a gap; depositing a dielectric material in the gap; and removing a portion of the dielectric material in the first S/D trench such that the first S/D trench extending through the dielectric material.

[0053] Another method of forming a semiconductor device comprises forming a sacrificial layer over a substrate; alternately depositing first semiconductor layers and second semiconductor layers over the sacrificial layer, wherein each of a material of the first semiconductor layers, a material of the second semiconductor layers, and a material of the sacrificial layer provides a different etching selectivity; patterning the first semiconductor layers, the second semiconductor layers, the sacrificial layer, and a top portion of the substrate to form a semiconductor stack; forming a first dummy gate structure and a second dummy gate structure over the semiconductor stack; etching along sidewalls of the first dummy gate structure and the second dummy gate structure to form a first source/drain (S/D) trench between the first dummy gate structure and the second dummy gate structure, and a second S/D trench along a sidewall of the second dummy gate structure away from the first dummy gate structure, wherein bottom surfaces of the first S/D trench and the second S/D trench are above the sacrificial layer; further etching the first S/D trench such that the bottom surface of the first S/D trench is below a bottom surface of the sacrificial layer, while keeping the second S/D trench unchanged; replacing the sacrificial layer with a dielectric layer; and epitaxially growing a first epitaxial feature in the first S/D trench and a second epitaxial feature in the second S/D trench.

[0054] In some embodiments, the method further comprises selectively removing the second semiconductor layers; replacing the first dummy gate structure with a first metal gate structure and replacing the second dummy gate structure a second metal gate structure, wherein the first and second metal gate structures wrap each of the first semiconductor layers; removing the substrate to expose the first epitaxial feature and the dielectric layer; and forming a conductive feature contacting the dielectric layer and the first epitaxial feature.

[0055] In some embodiments, the method further comprises forming an isolation feature around a lower portion of the semiconductor stack, wherein a top surface of the isolation feature is below a bottom surface of the sacrificial layer; wherein the removing the substrate includes removing a bottom portion of the substrate to expose the isolation feature; and removing the top portion of the substrate to form an opening in the isolation feature to expose the first epitaxial feature and the dielectric layer.

[0056] In some embodiments, the further etching the first S/D trench includes forming a patterned photoresist layer over the first dummy gate structure and the second dummy gate structure, wherein the patterned photoresist layer exposes the first S/D trench; and removing a portion of the sacrificial layer between the first dummy gate structure and the second dummy gate structure through the patterned photoresist layer. In some embodiments, a width of a removed portion of the sacrificial layer is more than about 10 nm.

[0057] The foregoing outlines features of several embodiments so that those skilled in the art may

better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a backside metal line extending lengthwise along a first direction; an isolation structure over the backside metal line; a contact via extend through the isolation structure and extending lengthwise along the first direction; a bottom dielectric layer over the contact via; a first stack of nanostructures disposed over the bottom dielectric layer; a second stack of nanostructures disposed over the bottom dielectric layer; a first gate structure wrapping around each of the first stack of nanostructures; a second gate structure wrapping around each of the second stack of nanostructures; and a source/drain feature disposed between the first stack of nanostructures and the second stack of nanostructures along the first direction, wherein the source/drain feature extends through the bottom dielectric layer.
2. The semiconductor structure of claim 1, wherein the contact via extends directly under the first gate structure and the second gate structure.
3. The semiconductor structure of claim 1, wherein the source/drain feature is electrically coupled to the contact via by way of a silicide layer.
4. The semiconductor structure of claim 1, wherein the contact via comprises W, Ru, Co, Cu, Mo, Ni, Ti, TiN, Ta, or TaN.
5. The semiconductor structure of claim 1, wherein the bottom dielectric layer comprises a thickness between about 6 nm and about 20 nm.
6. The semiconductor structure of claim 1, further comprising: an interconnect structure over the first gate structure, the second gate structure, and the source/drain feature.
7. The semiconductor structure of claim 1, wherein the backside metal line comprises Cu, Al, Co, W, Ti, Ta, or Ru.
8. The semiconductor structure of claim 1, further comprising: a contact etch stop layer disposed over the source/drain feature.
9. The semiconductor structure of claim 1, wherein the contact etch stop layer interfaces the isolation structure.
10. A semiconductor structure, comprising: a backside metal line extending lengthwise along a first direction; an isolation structure over the backside metal line; a contact via extending through the isolation structure and extending length wise along the first direction; a bottom dielectric layer over the contact via; a drain feature disposed on the bottom dielectric layer; a source/drain feature extending through the bottom dielectric layer to electrically coupled to the contact via; a stack of nanostructures extending lengthwise along the first direction between the drain feature and the source/drain; and a gate structure wrapping around each of the stack of nanostructures.
11. The semiconductor structure of claim 10, wherein the contact via is spaced apart from the gate structure and the drain feature by the bottom dielectric layer.
12. The semiconductor structure of claim 10, wherein the stack of nanostructures are interleaved by a plurality of inner spacer features, wherein the gate structure is spaced apart from the source feature by the plurality of inner spacer features.
13. The semiconductor structure of claim 10, wherein the source feature is electrically coupled to the contact via by way of a silicide layer.
14. The semiconductor structure of claim 13, wherein the silicide layer comprises TiSi, CoSi, or

MoSi.

15. The semiconductor structure of claim 10, wherein the bottom dielectric layer is sandwiched between a first gate spacer feature and a second gate spacer feature along a second direction perpendicular to the first direction.

16. The semiconductor structure of claim 15, wherein the first gate spacer and the second gate spacer are disposed over the isolation structure.

17. A method, comprising: depositing a sacrificial layer over a substrate; forming, over the sacrificial layer, a stack that includes a plurality of first semiconductor layers, and a plurality of second semiconductor layers interleaving the plurality of first semiconductor layers; patterning the stack, the sacrificial layer and a portion of the substrate to form a fin-shaped structure comprising a drain region, a source region, and a channel region disposed between the drain region and the source region; forming an isolation feature around the fin-shaped structure; forming a dummy gate stack over the channel region; forming a drain trench over the drain region, the drain trench terminating over the sacrificial layer; forming a source trench over the source region, the source trench extending through the sacrificial layer and into the substrate; replacing the sacrificial layer with a bottom dielectric layer; forming a source feature in the source trench and a drain feature in the drain trench; depositing an interlayer dielectric layer over the source feature and the drain feature; after the depositing of the interlayer dielectric layer, removing the dummy gate stack; selectively removing the plurality of second semiconductor layers in the channel region to release the plurality of first semiconductor layers as channel members; after the selectively removing of the plurality of second semiconductor layers, forming a gate structure to wrap around each of the channel members; thinning the substrate to expose the isolation feature; selectively removing the thinned substrate to form a via trench that exposes the bottom dielectric layer and the source feature; and forming a contact feature in the via trench, wherein the sacrificial layer and the plurality of second semiconductor layers comprise silicon germanium, wherein a germanium content of the sacrificial layer is smaller than a germanium content of the plurality of second semiconductor layers.

18. The method of claim 17, further comprising: after the forming of the contact feature in the via trench, performing an anneal process to form a silicide layer between the contact via and the source feature.

19. The method of claim 17, wherein the germanium content of the sacrificial layer is between about 10% and about 35%, wherein the germanium content of the plurality of second semiconductor layers is between about 20% and about 50%.

20. The method of claim 17, wherein the replacing of the sacrificial layer comprises: selectively removing the sacrificial layer; after the selectively removing, depositing a dielectric material over the source trench; and after the depositing of the dielectric material, anisotropically etching the dielectric material in the source trench to expose the substrate.
